Toward f: a general factor of human flourishing

A dominant general factor underlies virtually all major wellbeing measurement instruments, with bifactor models consistently explaining 40–60% of common variance across psychological wellbeing, mental health continuum, PERMA, and character strength assessments. This factor — which we formally designate **f** — is analogous to Spearman's g for intelligence and Caspi's p for psychopathology, PubMed +2 but its ontological status is fundamentally different from both. Where g indexes a performance capacity and p indexes dysfunction liability, f indexes something closer to Aristotle's eudaimonia: the coherent, self-reinforcing activity of a well-functioning human life. The contribution here is a formal mathematical specification of f that reconciles three traditions that rarely speak to one another — psychometric factor analysis, the capabilities approach from welfare economics, and dynamical systems theory — while remaining honest about the deep philosophical tensions this synthesis creates.

The empirical positive manifold is overwhelming

Across every major wellbeing measurement tradition, the same pattern recurs: nominally distinct dimensions correlate so strongly that a single general factor dominates the variance structure. In Ryff's Psychological Well-Being model, Springer and Hauser found latent factor correlations reaching **.954** between purpose in life and self-acceptance, with four of six dimensions effectively indistinguishable. ScienceDirect The Mental Health Continuum Short Form shows omega hierarchical values of approximately **.80** for a general wellbeing factor, PubMed Central +2 with explained common variance (ECV) of .82 in some samples ResearchGate — high enough that separate subscale scoring is "not theoretically justified." PubMed Central When Goodman and colleagues administered both PERMA and subjective wellbeing instruments and extracted bifactor models, the two general factors correlated at **.98**, suggesting near-total construct redundancy. Taylor & Francis Online The VIA character strengths yield a "good character" general factor that, once modeled, renders almost all 24 specific strengths non-predictive of life satisfaction. VIA Institute +2

These findings converge across instruments, populations, and analytic methods. Bifactor exploratory structural equation modeling (bifactor ESEM) consistently provides the best fit, Frontiers PubMed whether applied to Ryff's 54-item scale, ResearchGate Keyes's 14-item MHC-SF across four nations,

ResearchGate PubMed or the VIA across 447,573 participants. Springer Academia.edu The pattern is not merely statistical: Bjørndal and colleagues demonstrated that the general wellbeing factor (which they label "h") shows the **highest heritability** (26–40%) among all wellbeing dimensions in genetically informative twin samples ResearchGate +2 — precisely the pattern one expects if the general factor captures something real about individual variation rather than method artifact. Disabato and colleagues' Hierarchical Framework of Wellbeing (HiFWB) now proposes a four-level hierarchy — general h at the apex, lenses (subjective, psychological, social wellbeing), contents (affects, cognitions, virtues), and specific characteristics — directly mirroring Carroll's three-stratum model of intelligence. Frontiers PubMed Central

The three wellbeing dimensions that most consistently retain specificity after extracting the general factor are **autonomy** (in Ryff's model), **social wellbeing** (in Keyes's model), and **engagement** (in PERMA). This suggests the general factor primarily captures a positive self-evaluative orientation — environmental mastery, self-acceptance, purpose, meaning, positive emotion, and life satisfaction load most heavily — while autonomy, social functioning, and flow-state engagement represent partially distinct processes.

What f is: philosophical architecture of the construct

The most consequential theoretical decision in defining f is whether its indicators are **reflective** (caused by f) or **formative** (constitutive of f). This is not a technical measurement question but a philosophical one about the nature of flourishing itself.

Aristotle's eudaimonia provides the original framework. The *ergon* argument holds that humans flourish by exercising their characteristic function — rational activity in accordance with virtue — over a complete life. Critically, virtues are **constitutive elements** of eudaimonia, not merely instrumental causes. Courage, justice, and practical wisdom don't produce flourishing as an external effect; they partly compose it. This constitutive relationship maps naturally to a formative measurement model, where dropping an indicator (say, close relationships) changes the construct's meaning rather than merely reducing measurement precision.

Yet the empirical evidence stubbornly favors reflective structure. The positive manifold — all wellbeing indicators positively intercorrelating — and the dominance of a single factor are hallmarks of a reflective latent variable. ScienceDirect Here, the resolution proposed by the dynamical systems literature is decisive.

Van der Maas and colleagues proved formally that a **mutualism model** — in which cognitive processes mutually reinforce each other through development — produces data perfectly fitting a single-factor model even though no latent variable generated the data. (ResearchGate +3) The "general factor" is an emergent equilibrium property of the coupled system, not a common cause.

This dissolves the constitutive-versus-causal dilemma. Under a mutualism interpretation, f is neither a hidden cause producing observable wellbeing dimensions nor a mere summary composite. It is the **emergent coherence of a self-reinforcing system** — the degree to which a person's wellbeing dimensions have reached a mutually reinforcing equilibrium. The components are indeed constitutive (they compose the system), and the general factor is indeed real (it captures genuine structural properties of that system), but the causal arrows run laterally between components rather than vertically from a latent variable. Alexandrova's concern that factor analysis substitutes statistical pattern-finding for philosophical argumentation is thus addressed: (Notre Dame Philosophical Revi…) the statistical pattern reflects a real dynamical process, but one whose proper description is a network, not a latent cause.

The "thick concept" problem identified by Bernard Williams also finds resolution here. Flourishing carries irreducible evaluative content — to call someone flourishing is simultaneously to describe their state and to commend it. A purely formal f cannot encode this normative dimension. The specification below therefore explicitly distinguishes the **descriptive-structural** component of f (the coupling strength and equilibrium quality of the wellbeing system, which is mathematically formalizable) from the **evaluative** component (the judgment that this system state constitutes a good human life, which requires philosophical argument external to the model). The model captures the former; the latter must be supplied by a substantive theory of the good — whether Aristotelian, capability-based, or hybrid.

Formal mathematical specification of f

Model 1: standard bifactor CFA

Let y_{ij} denote person i's score on indicator j ($j = 1, \ldots, J$), where indicators are grouped into K domains (e.g., positive emotion, engagement, relationships, meaning, accomplishment, health). The bifactor specification is:

$$y_{ij} = \mu_j + \lambda_j^{(f)} f_i + \lambda_j^{(s_k)} s_{ki} + \epsilon_{ij}$$

where f_i is the general flourishing factor for person i, s_{ki} is the specific factor for domain k to which item j belongs, $\lambda_j^{(f)}$ and $\lambda_j^{(s_k)}$ are factor loadings, and $\epsilon_{ij} \sim N(0, \theta_j)$. By bifactor construction, $f \perp s_k$ for all k, and $s_k \perp s_{k'}$ for $k \neq k'$. (PubMed Central +3)

The proportion of systematic variance attributable to f is indexed by **omega hierarchical**: $\omega_h = (\sum_j \lambda_j^{(f)})^2 / [\sum_j (\lambda_j^{(f)})^2 + \sum_k \sum_{j \in k} (\lambda_j^{(s_k)})^2 + \sum_j \theta_j]$. Empirical estimates across major instruments converge on $\omega_h \approx .71$–$.86$, indicating that **approximately 75–85% of reliable variance is attributable to f**. (PubMed Central)

This model represents what the data look like, but says nothing about what generates the pattern.

Model 2: the mutualism specification (the generative model)

Replace the latent-cause assumption with a dynamical system. Let $x_i(t) = (x_{i1}(t), \ldots, x_{iK}(t))^\top$ represent person i's state vector across K wellbeing domains at time t. The mutualism model, adapted from van der Maas et al. (2006), specifies coupled logistic growth:

$$\frac{dx_{ik}}{dt} = a_{ik}\, x_{ik} \left(1 - \frac{x_{ik}}{C_{ik}} + \sum_{l \neq k} \frac{M_{kl}^{(i)}\, x_{il}}{C_{ik}} \right) + \sigma_{ik}\, \xi_{ik}(t)$$

where $a_{ik} > 0$ is the intrinsic growth rate of domain k for person i, C_{ik} is the carrying capacity (the domain's asymptotic level absent mutualistic support), $M_{kl}^{(i)} \geq 0$ is the mutualistic coupling strength between domains k and l, and $\xi_{ik}(t)$ is white noise.

At equilibrium ($dx_{ik}/dt = 0$), each domain's level exceeds its individual carrying capacity because mutual facilitation allows domains to "lift" each other: $x_{ik}^* > C_{ik}$ when $M_{kl} > 0$. Individual differences in the equilibrium vector x_i^* arise from variation in parameters $(C_{ik}, a_{ik}, M_{kl}^{(i)})$. The covariance matrix $\mathrm{Cov}(x^*)$ across individuals exhibits a positive manifold — all positive off-diagonal

entries — even though no single latent variable generated the data. :antCitation[]{citations="52ee9130-080f-4f16-baab-a4fd1d02c40d"} Factor analysis of this covariance matrix yields a dominant first eigenvalue, :antCitation[]{citations="7e29626f-14a1-4df1-bc82-6c9c996432ce,7d0bea40-4dbb-4493-9cbf-5ec4c42fdf9d"} which is the **psychometric f**.

Definition: The general factor of flourishing f is formally defined as the **first principal component of the equilibrium covariance matrix** of a positively coupled wellbeing system. Its magnitude for a given individual reflects the joint influence of (a) the person's carrying capacities C_{ik} across domains, (b) the strength of mutualistic coupling $M_{kl}^{(i)}$, and (c) the developmental history through which the system reached equilibrium.

The **mean coupling strength** $\bar{M}_i = \frac{2}{K(K-1)} \sum_{k<l} M_{kl}^{(i)}$ governs how coherently the system behaves. Higher $\bar{M}$ produces stronger positive manifold, greater dominance of f in the factor structure, and — crucially — deeper basins of attraction around the flourishing equilibrium.

Model 3: Lyapunov stability and resilience

The connection to resilience is made precise through the Jacobian. At equilibrium x^*, linearize the system: $\frac{d\delta x}{dt} = J\,\delta x$, where $J_{kl} = \partial \dot{x}_k / \partial x_l \big|_{x^*}$ is the Jacobian matrix. Local asymptotic stability requires all eigenvalues of J to have negative real parts: $\mathrm{Re}(\lambda_k) < 0$ for all k.

The **dominant eigenvalue** λ_1 (the eigenvalue closest to zero) governs the system's recovery rate from perturbation. The characteristic return time is $\tau = 1/|\mathrm{Re}(\lambda_1)|$. As $\lambda_1 \to 0$, the system exhibits **critical slowing down** — increased autocorrelation, variance, and cross-correlation in time-series data (PubMed Central) — which van de Leemput and colleagues demonstrated empirically as an early warning signal for depression onset. (ADS) (Tilburg University)

Define **resilience** as the depth of the basin of attraction around the flourishing equilibrium x^*: $\mathcal{R}(x^*) = \inf\{\|\delta x\| : x^* + \delta x \notin \mathcal{B}(x^*)\}$, where $\mathcal{B}(x^*)$ is the basin of attraction. A Lyapunov function $V(x)$ with $V(x^*) = 0$, $V(x) > 0$ for $x \neq x^*$, and $\dot{V} \leq 0$ certifies stability; the sublevel set $\{x : V(x) \leq c\}$ provides a conservative inner approximation of $\mathcal{B}(x^*)$.

For the mutualism system, a natural candidate Lyapunov function is:

$$V(\delta x) = \delta x^\top P \, \delta x$$

where P is the positive-definite solution to the Lyapunov equation $J^\top P + PJ = -Q$ for some positive-definite Q. This quadratic form defines an ellipsoidal approximation to the basin of attraction. The system's resilience is thus characterized by the shape and size of this ellipsoid.

The key theoretical link: Stronger mean coupling $\bar{M}$ simultaneously produces (a) a more dominant psychometric f (stronger positive manifold), (b) higher equilibrium wellbeing levels x^*, and (c) deeper basins of attraction (greater resilience) — but only up to a point. Excessively strong coupling can destabilize the system, as Borsboom's network theory of psychopathology demonstrates: strongly connected symptom networks can become trapped in disorder states. Frontiers +3 There exists an **optimal coupling regime** where f is maximized and resilience is greatest.

Model 4: mapping the capabilities approach onto the factor model

Sen's capability approach formalizes a person's capability set as $Q_i = \{b_i \mid b_i = f_i(c(x_i)),\ f_i \in F_i,\ x_i \in X_i\}$ where $b_i \in \mathbb{R}^K$ is a functioning vector, $c(\cdot)$ converts commodities to characteristics, and F_i is the set of feasible utilization functions. Uchicago The chosen functioning vector $b_i^* \in Q_i$ represents achieved functionings.

The mapping to the factor model is:

- **Functionings** b_i^* correspond to the observed wellbeing domain scores $(x_{i1}^*, \ldots, x_{iK}^*)$ at equilibrium — the realized values that enter factor analysis.
- **Capabilities** Q_i correspond to the **support of the distribution** of potential equilibria — the set of functioning vectors achievable given person i's parameters (C_{ik}, a_{ik}, M_{kl}) and environmental constraints. Two persons may have identical current functioning vectors b^* but radically different capability sets Q if one faces severe constraints and the other does not.
- **Nussbaum's threshold** maps to a minimum-value constraint on each domain: $x_{ik}^* \geq z_k$ for all k, where z_k is the threshold for domain k. This is explicitly non-compensatory — high meaning

cannot offset deprivation in bodily health — and corresponds to a conjunction rather than a weighted sum. (D-Scholarship@Pitt)

Formally, define a **capability-adjusted flourishing measure**:

$$f_i^{\text{cap}} = f(x_i^*) \cdot 1\left[\min_k x_{ik}^* \geq z_k\right] \cdot h(Q_i)$$

where $f(x_i^*)$ is the standard psychometric f (the general factor score), $1[\cdot]$ is Nussbaum's threshold indicator (zero if any domain falls below threshold), and $h(Q_i)$ is a measure of the "size" of the capability set (e.g., the Lebesgue measure of Q_i, or the expected value of f over the distribution of achievable equilibria). This three-part structure captures: the quality of the current attractor state, the non-compensatory minimum requirement, and the freedom to have achieved otherwise.

The Alkire-Foster multidimensional poverty index provides the measurement-level bridge. Their deprivation score $c_i = \sum_j w_j g_{ij}^0$ (with $g_{ij}^0 = 1$ if $x_{ij} < z_j$) and dual cutoff identification ($\rho_k = 1$ if $c_i \geq k$) operationalize the threshold logic for population-level assessment. The adjusted headcount ratio $M_0 = H \times A$ is decomposable by dimension, preserving the multidimensional information that a single f score collapses. (ophi)

How f relates to p: not simple opposites

The relationship between the general factor of flourishing and the general factor of psychopathology is more nuanced than a bipolar spectrum. In general populations, the correlation between wellbeing and psychopathology factors is moderate: typically **r ≈ −.40 to −.60**. (PubMed Central) Keyes's dual continua model, (Arabpsychology) supported by roughly 87.5% of studies in systematic reviews, holds that these are distinct dimensions: (PubMed Central) (PLOS) a person can have low psychopathology but also low wellbeing (the "languishing" state), or high wellbeing alongside some psychopathology symptoms. (PsyToolkit)

However, in clinical populations, the distinction collapses. Franken and colleagues found correlations of −.72 to −.88 between MHC-SF and psychopathology in patients with personality and mood disorders.

PubMed Central Chokhani and colleagues' bifactor model incorporating both wellbeing and psychopathology items in 15,258 adolescents found that a general factor captured **47% of common variance**, with wellbeing items loading negatively — consistent with a single underlying mental health dimension. PLOS

The dynamical systems framework resolves this apparent contradiction. In the coupled system, flourishing and disorder represent **alternative stable states** separated by a tipping point. For individuals whose system parameters place them far from the bifurcation (in general populations), the two attractors are well-separated and appear as distinct dimensions. For individuals near the bifurcation (clinical populations), the basins of attraction overlap and the distinction dissolves. The moderate correlation in general populations reflects the fact that parameters producing deeper flourishing basins (stronger positive coupling, higher carrying capacities) also tend to produce shallower disorder basins — but the relationship is not deterministic because the attractors have partially independent parameter dependencies.

This explains why critical slowing down — increased autocorrelation, variance, and cross-correlation in emotional time series — predicts both the onset and termination of depression. As the system approaches the bifurcation between flourishing and disorder, recovery from perturbation slows regardless of which direction the transition will go. **The dominant Jacobian eigenvalue λ_1 approaching zero is the common signature, whether the person is about to fall from flourishing into disorder or recover from disorder into flourishing.**

Five domains with a hierarchical architecture

VanderWeele's Harvard Human Flourishing Program provides the most carefully justified domain specification. His five core domains — **(1) happiness and life satisfaction, (2) mental and physical health, (3) meaning and purpose, (4) character and virtue, (5) close social relationships** — plus the sustaining domain of **(6) financial and material stability**, were selected because each constitutes "an end in itself" rather than merely an instrumental means. Harvard PNAS The Flourish Index (10 items, $\alpha = .89$) and Secure Flourish Index (12 items, $\alpha = .86$) have been validated across the United States, China, Sri Lanka, Cambodia, and Mexico, Harvard and deployed in the Global Flourishing Study across 200,000+ participants in 22 countries.

These domains map onto the bifactor structure as follows: the general factor f captures the shared positive-evaluative orientation across all domains (Lhbikos) (primarily driven by self-acceptance, life satisfaction, meaning, and positive emotion), while specific factors capture domain-unique processes. (PubMed Central) The three most independent specific factors — **autonomy/agency**, **social functioning**, and **engagement/flow** — represent processes that contribute to flourishing but operate through partially distinct mechanisms. This parallels the intelligence hierarchy where g captures shared cognitive efficiency while specific factors (verbal, spatial, processing speed) capture distinct abilities.

The mutualism model predicts exactly which domain pairs should show the strongest coupling. Relationships enhance meaning (affiliative connection provides purpose); meaning enhances engagement (purpose motivates absorption in valued activities); engagement enhances accomplishment (flow states produce mastery); accomplishment enhances self-acceptance (achieving goals validates self-worth); and self-acceptance enhances positive emotion (self-regard generates hedonic benefit). These reciprocal pathways — documented extensively in longitudinal research — are the mutualistic interactions M_{kl} whose collective strength produces f as an emergent property.

Conclusion: what f reveals and what it cannot

The formal specification of f advanced here makes three novel contributions. First, it resolves the constitutive-causal impasse by grounding f in a mutualism model where components constitute the system and the general factor emerges from their dynamic interaction — neither a hidden cause nor a mere composite. Second, it provides the first explicit mapping between the capabilities approach and psychometric factor models, showing that functionings are realized equilibria, capabilities are the support of potential equilibria, and Nussbaum's threshold logic corresponds to a non-compensatory constraint on the domain state vector. Third, it connects the static cross-sectional factor structure to Lyapunov stability theory, identifying mean coupling strength as the parameter that simultaneously governs factor dominance, equilibrium wellbeing, and resilience — with the critical prediction that there exists an optimal coupling regime beyond which the system becomes vulnerable to cascading failures.

What f cannot do is equally important. It cannot settle the evaluative question of whether any particular equilibrium state constitutes a good life — that requires philosophical argument, not factor analysis. It cannot eliminate the tension between compensatory aggregation (which the psychometric model performs) and non-compensatory threshold requirements (which justice demands). And it cannot

guarantee cross-cultural universality: the factor structure has been validated primarily in WEIRD samples, and whether the same mutualistic coupling architecture holds in collectivist, low-income, or non-Western contexts remains an open empirical question. ResearchGate +3 The model provides a formal language for investigating these questions — not definitive answers to them.